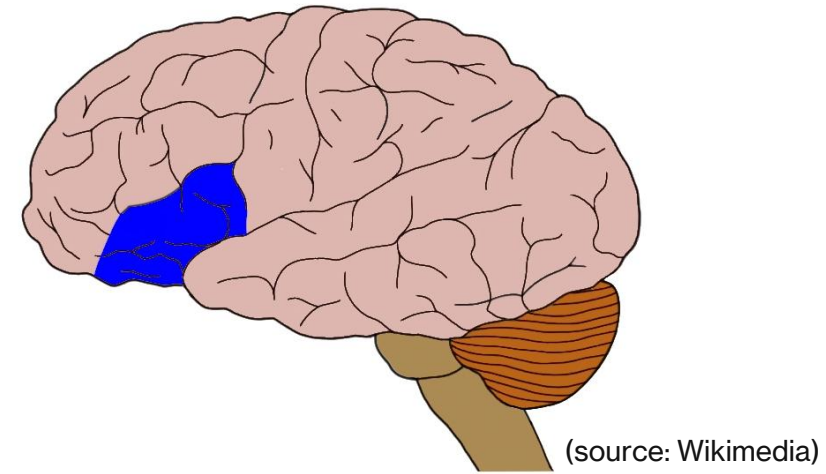


Attention-guided encoding models uncover syntactic computations in the in the inferior frontal gyrus during speech comprehension

Gwillams Lab Meeting 4/2/2026

The Inferior Frontal Gyrus (IFG) and its role in speech comprehension



- Highly dynamic and structurally complex
 - Plays roles in speech comprehension, production, working memory, and more (Fedorenko et al. 2012)
- In terms of composition plays important roles in both syntactic and semantic processing (Ishkhanyan et al. 2020, Klaus & Hartwigsen 2019)
- Traditional investigations use structured experimental regimes (contrast fMRI/TMS)
 - Naturalistic environments have unique strengths, diversity in language structures and long-term dependencies

Data

Microelectrode “Utah Array” data

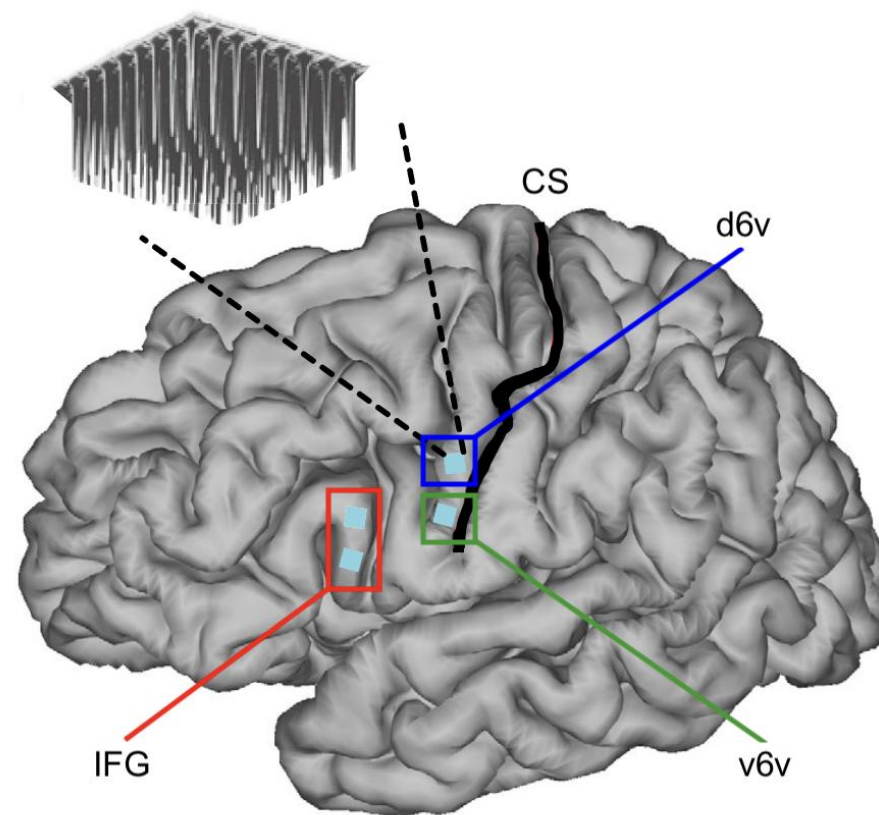
2 electrode arrays, 128 electrodes in total

Naturalistic Story Stimuli from *The Moth Radio Hour*

T12 listened to ~72 stories across 8 sessions

Resampled to 100Hz and epoched on word onset (± 2.3 s)

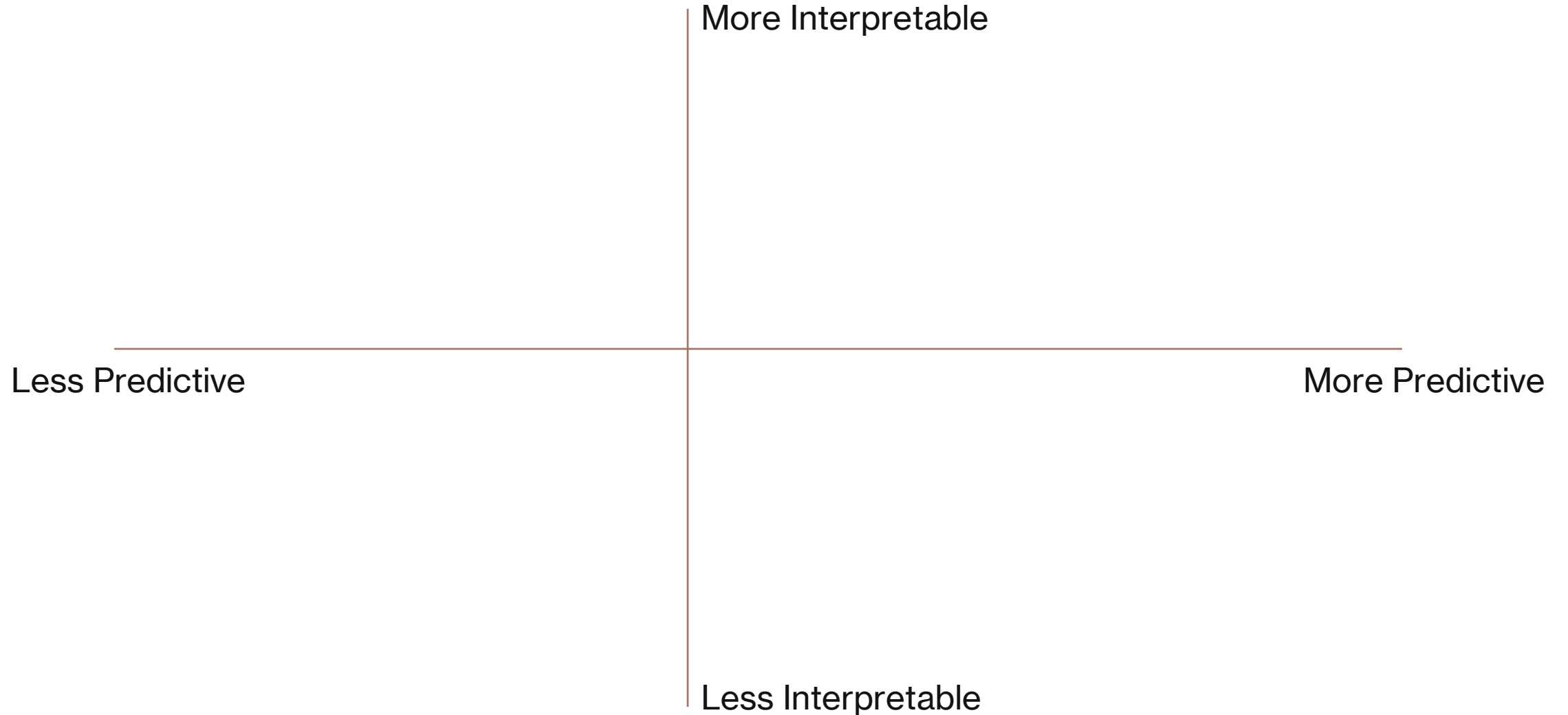
0.21s sliding average applied



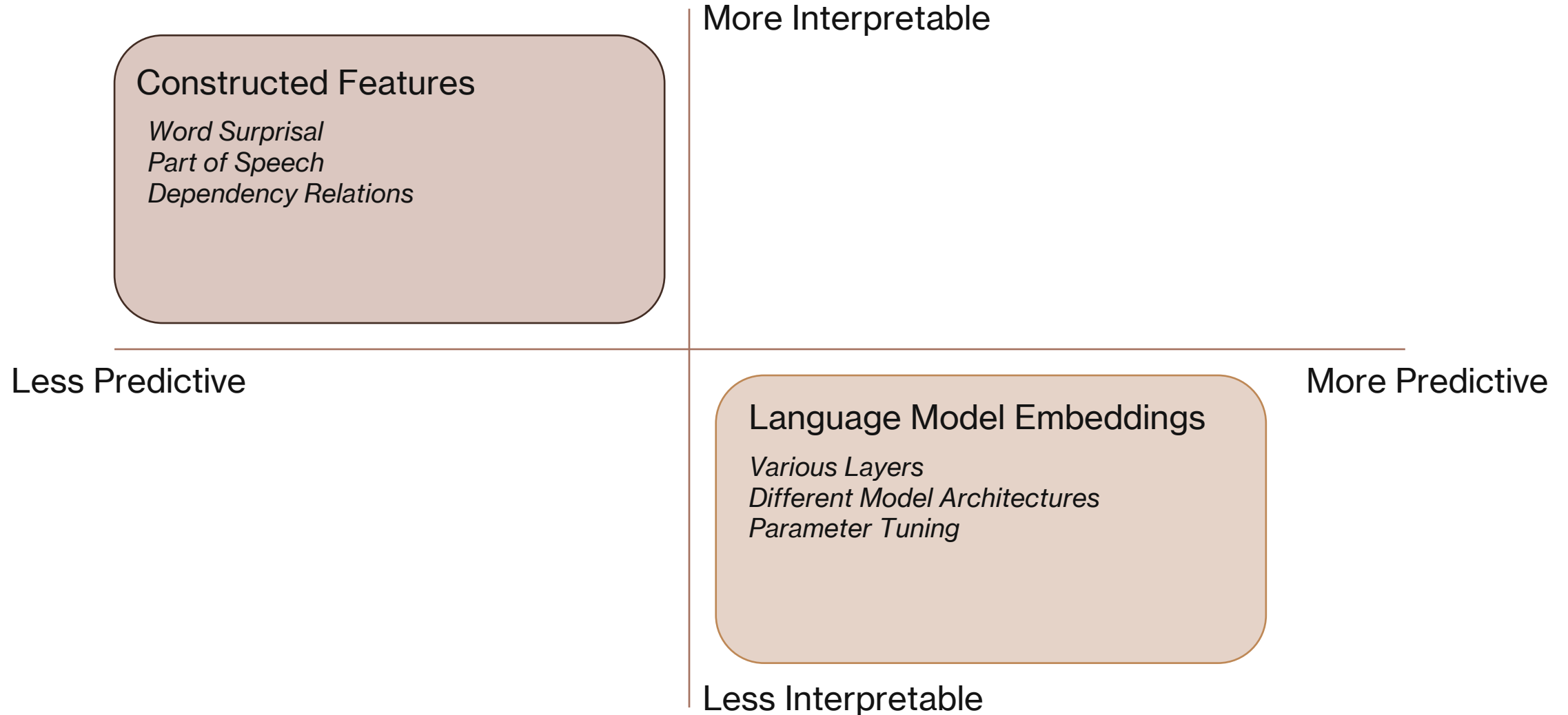
“I was three months pregnant with identical twin boys when my husband, Ross, and I learned that one of them had a fatal birth defect. Our son Thomas had anencephaly, which ...”



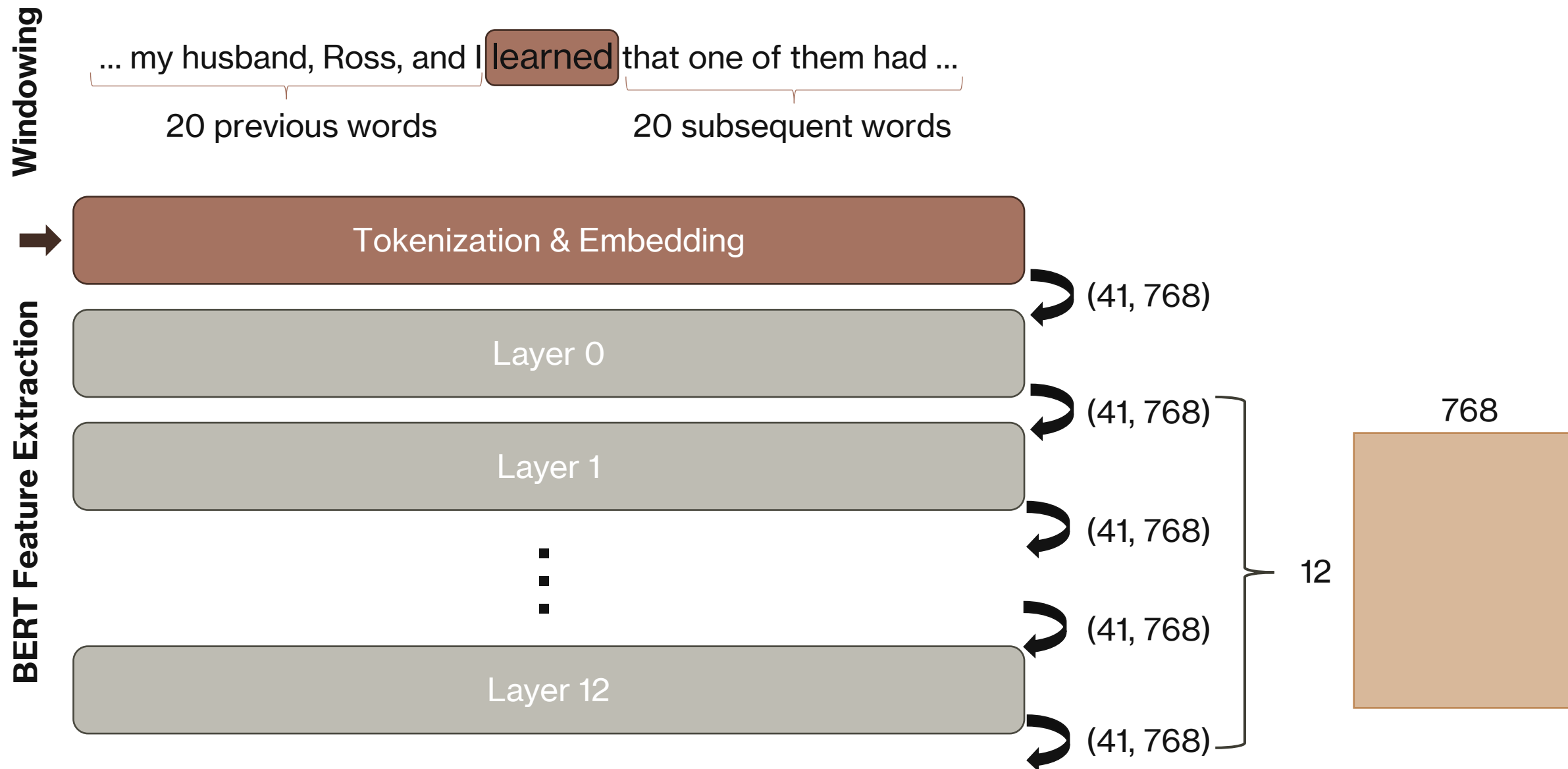
Designing Encoding Models



Designing Encoding Models



Feature Engineering



Feature Engineering

Windowing

... my husband, Ross, and I learned **that** one of them had ...

20 previous words

20 subsequent words

Tokenization & Embedding

(41, 768)

Layer 0

(41, 768)

Layer 1

(41, 768)

⋮

(41, 768)

Layer 12

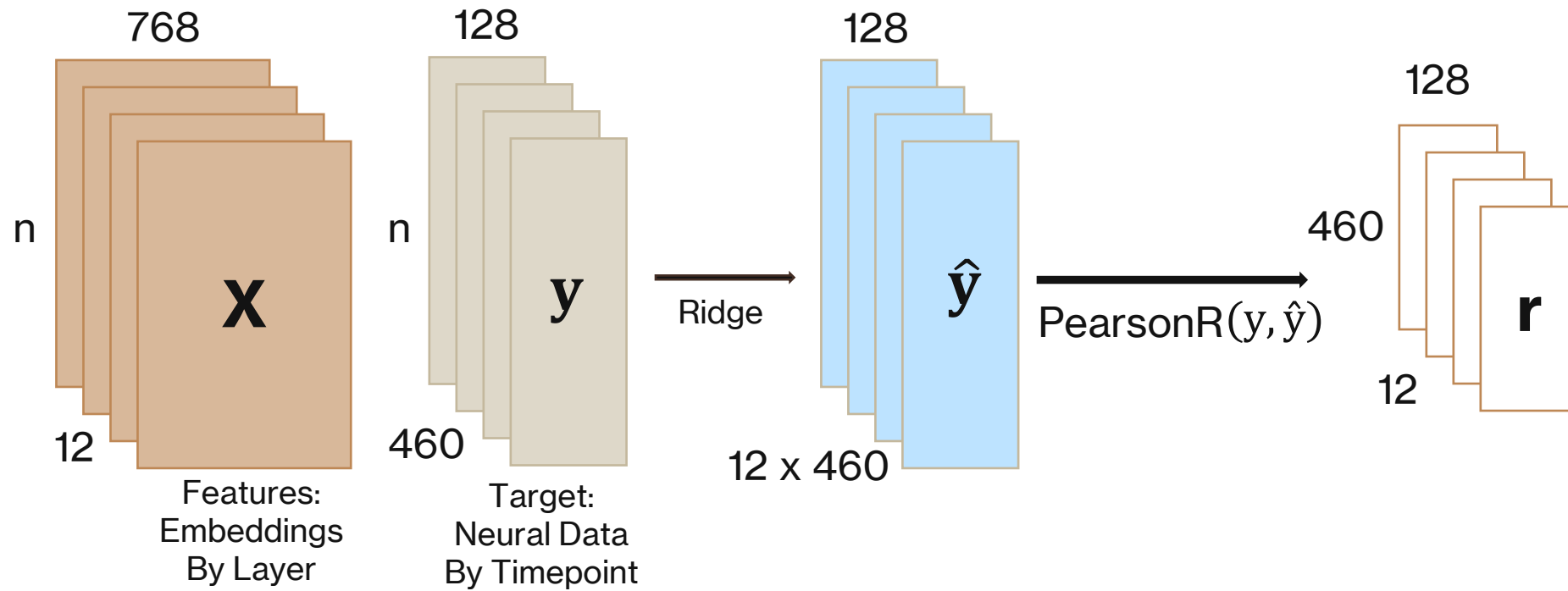
(41, 768)

12

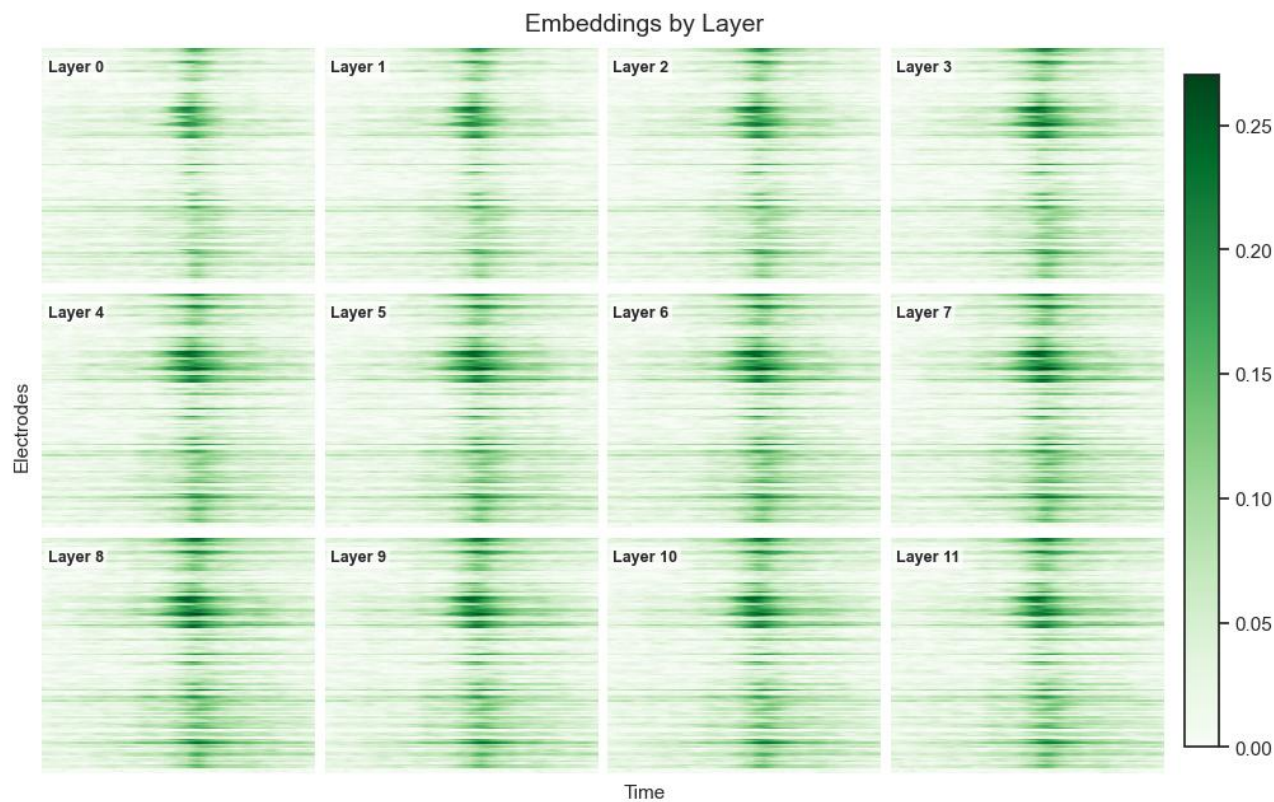
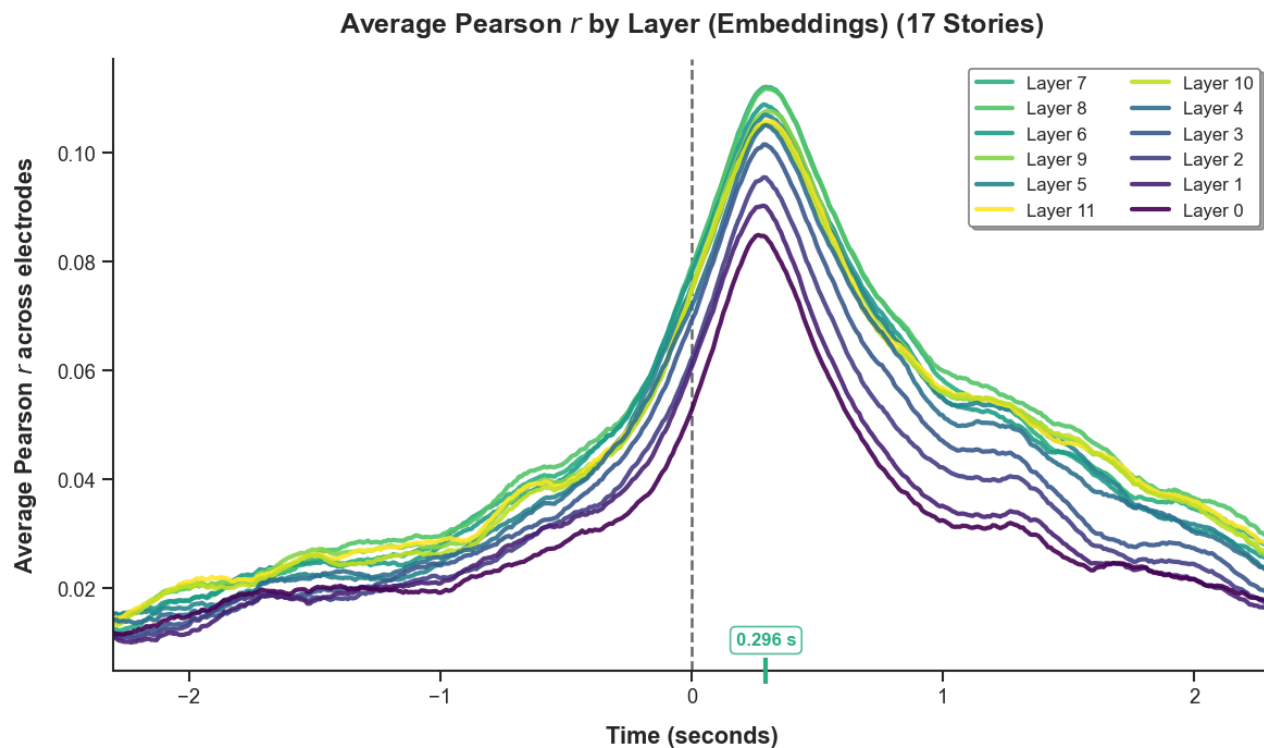
samples

768

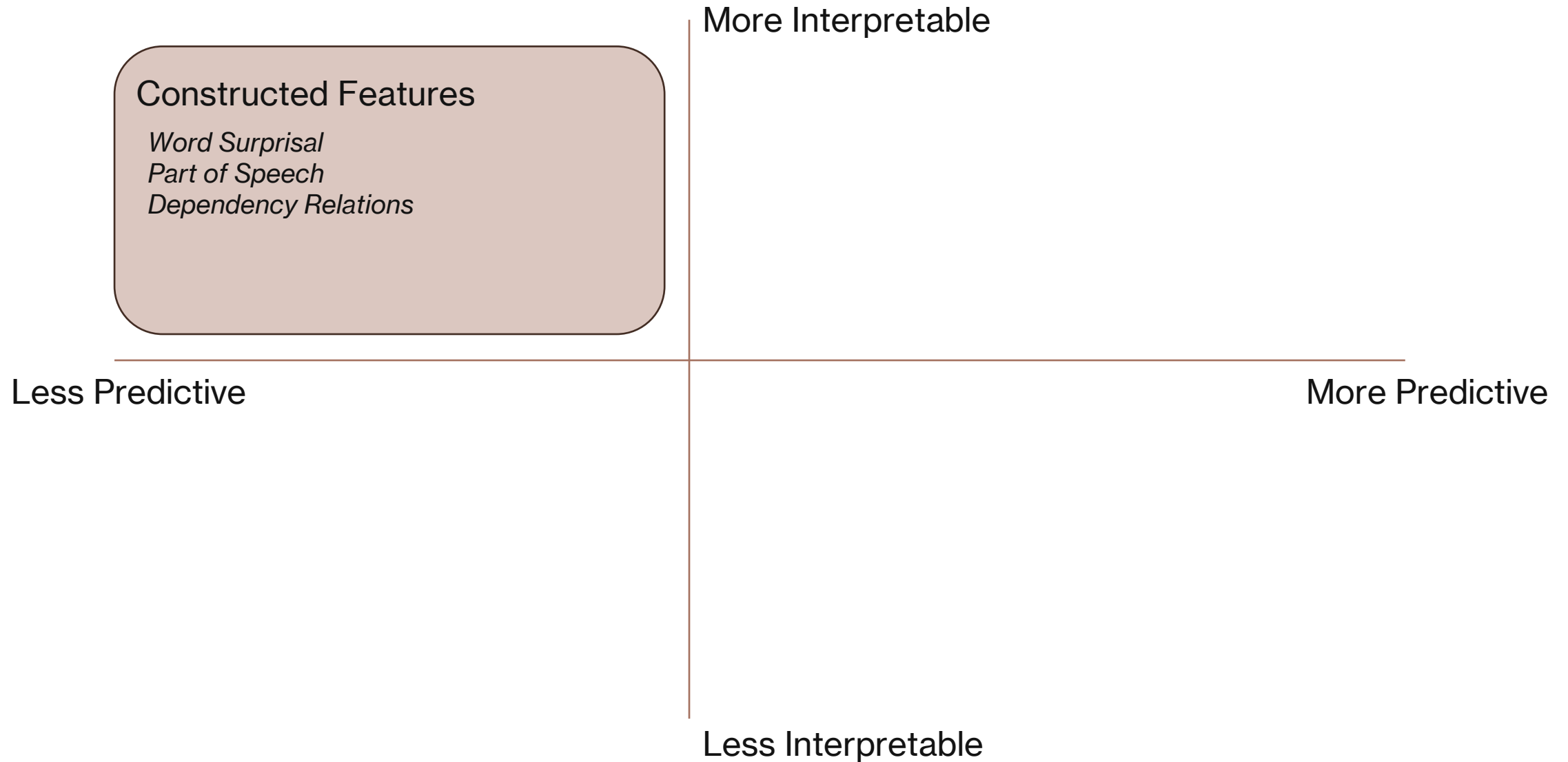
Embedding Encoding Model



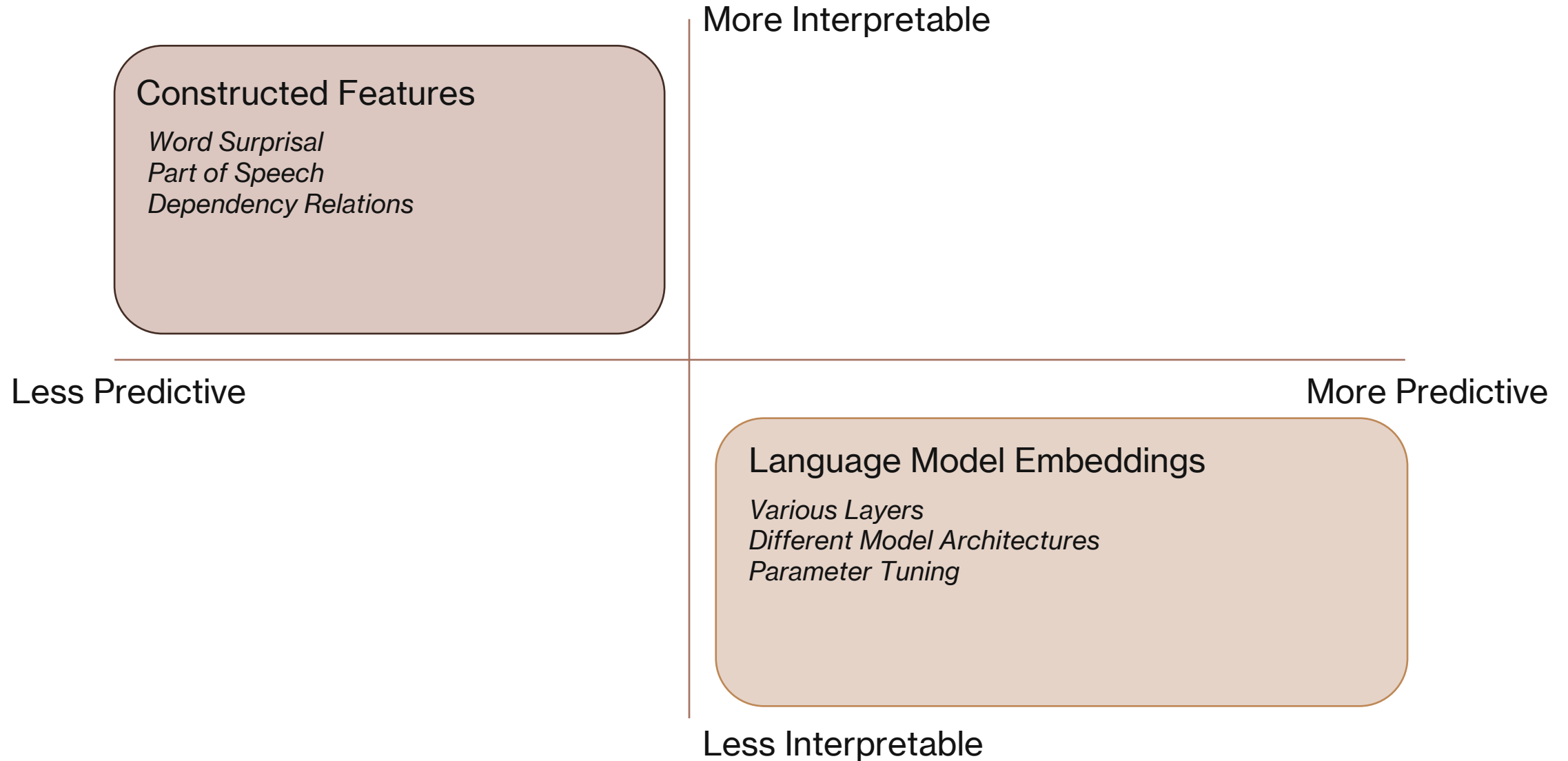
BERT Embedding Performance



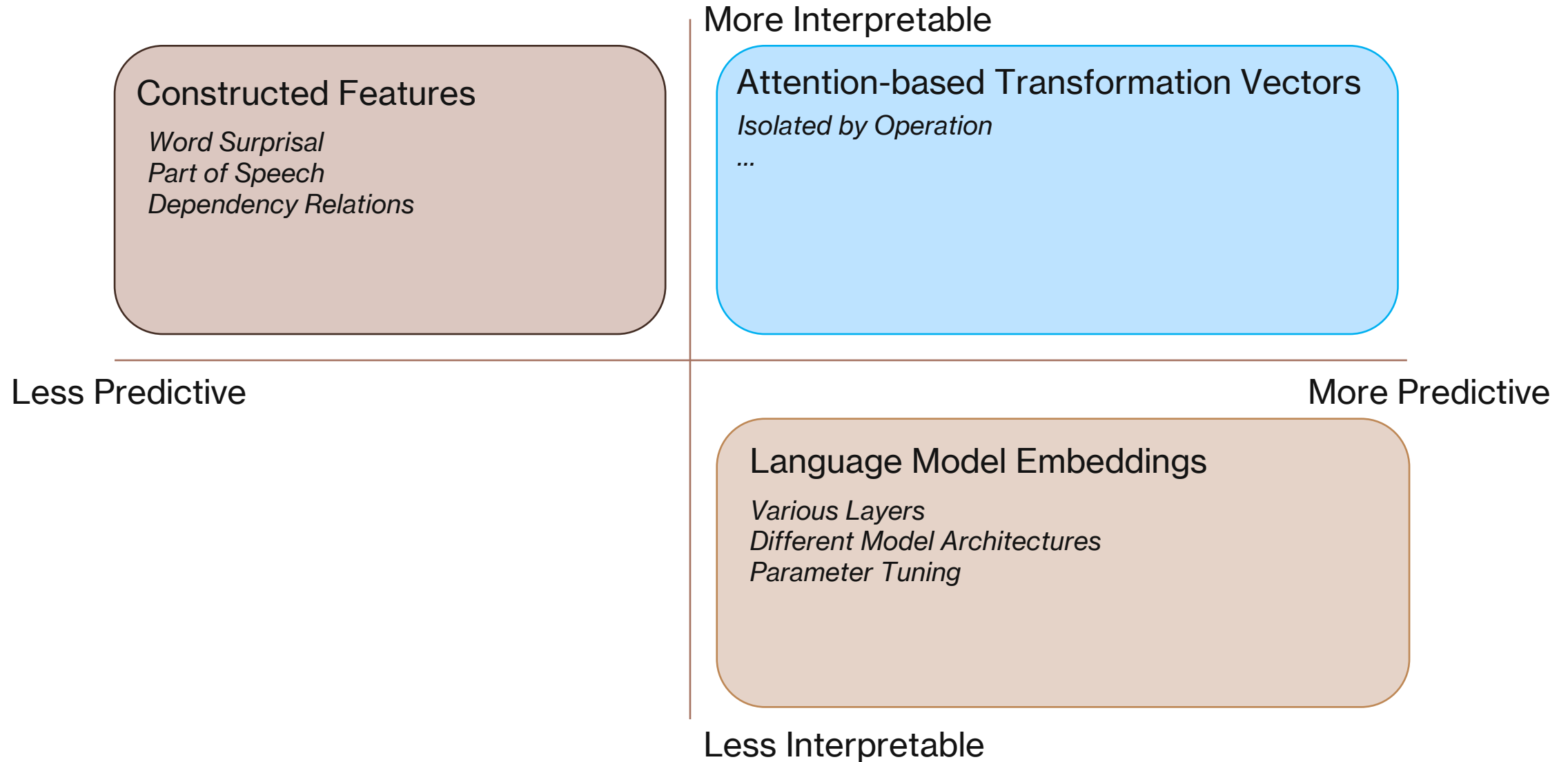
Designing Encoding Models



Designing Encoding Models



Designing Encoding Models



BERT Layer

BERT Layer

Input Embedding

768 by # of Tokens



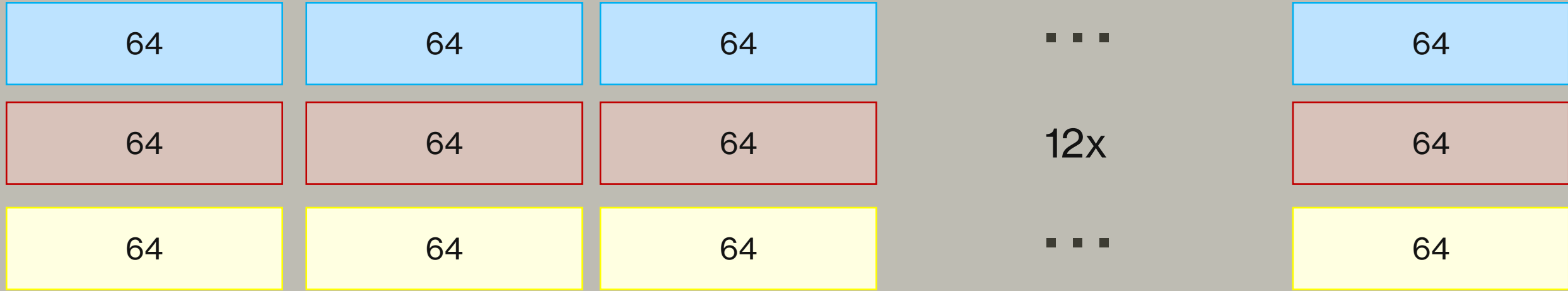
Learned Weight Matrices

Value Tensor

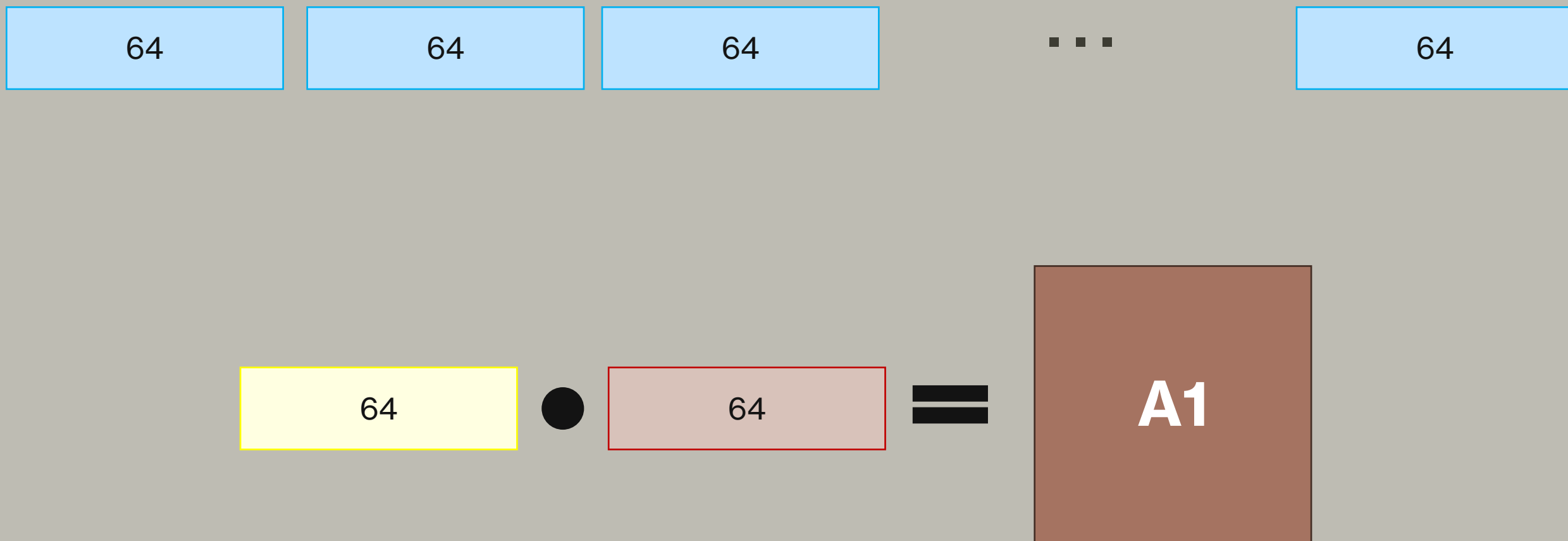
Key Tensor

Query Tensor

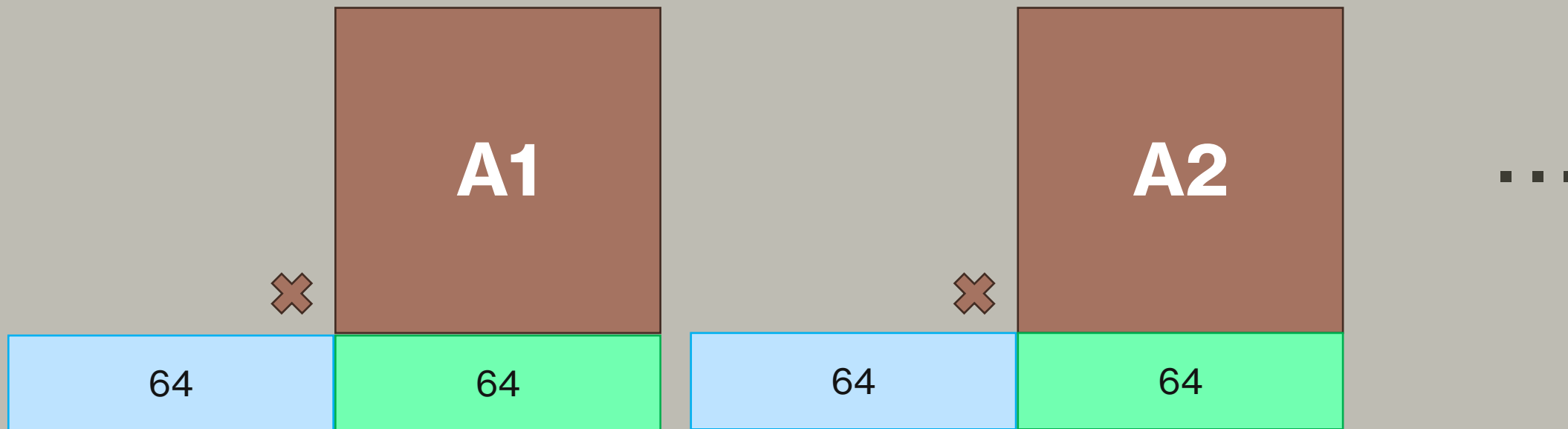
BERT Layer



BERT Layer



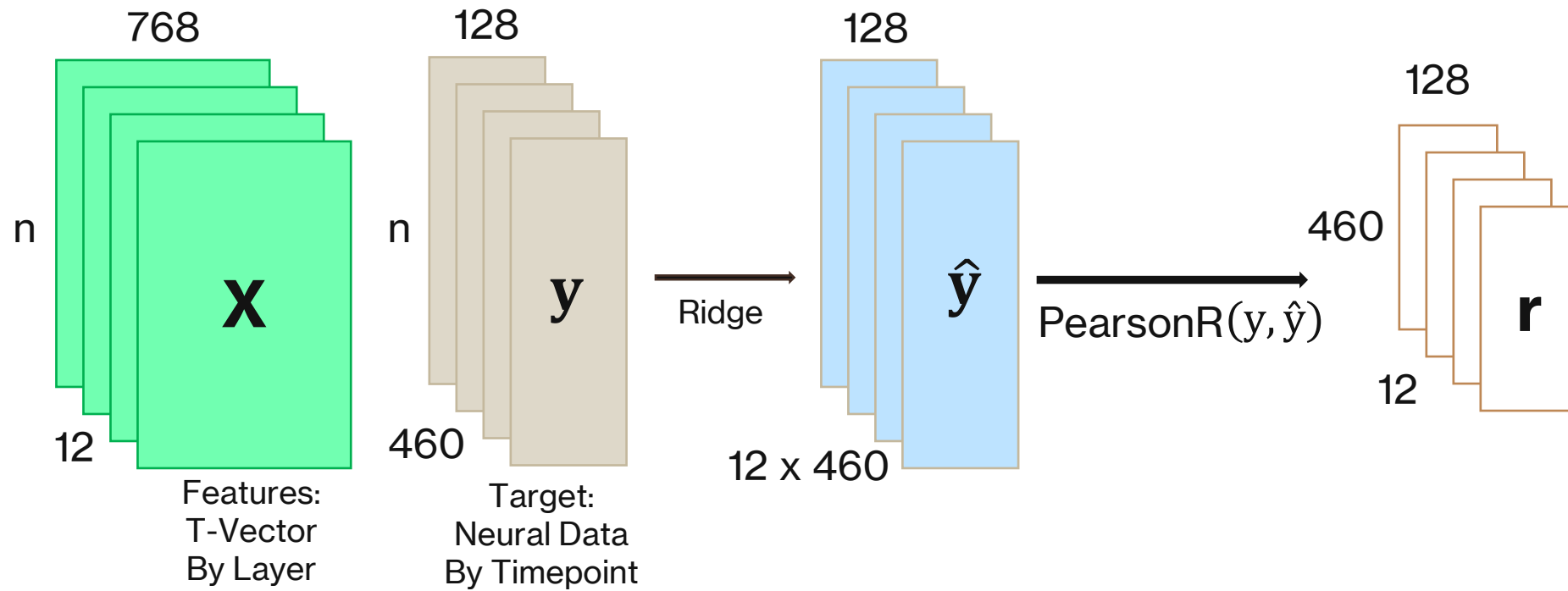
BERT Layer



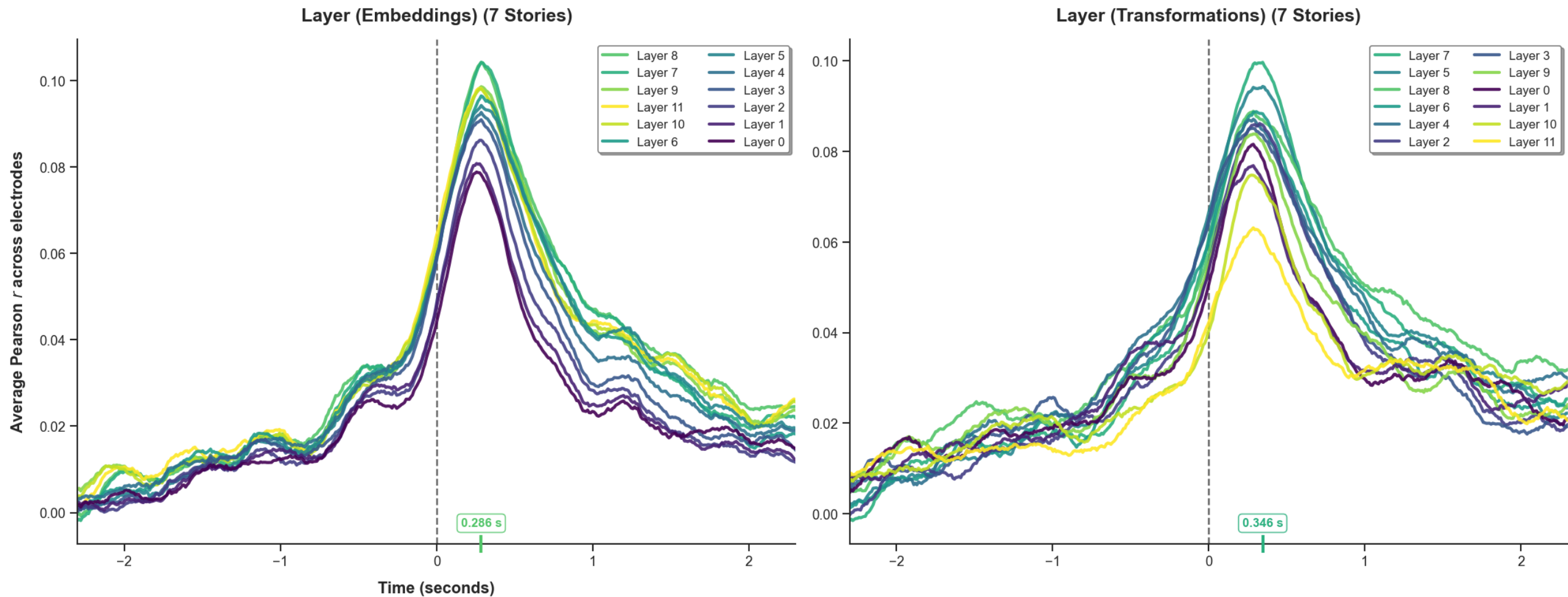
Transformation Vector

768 by # of Tokens

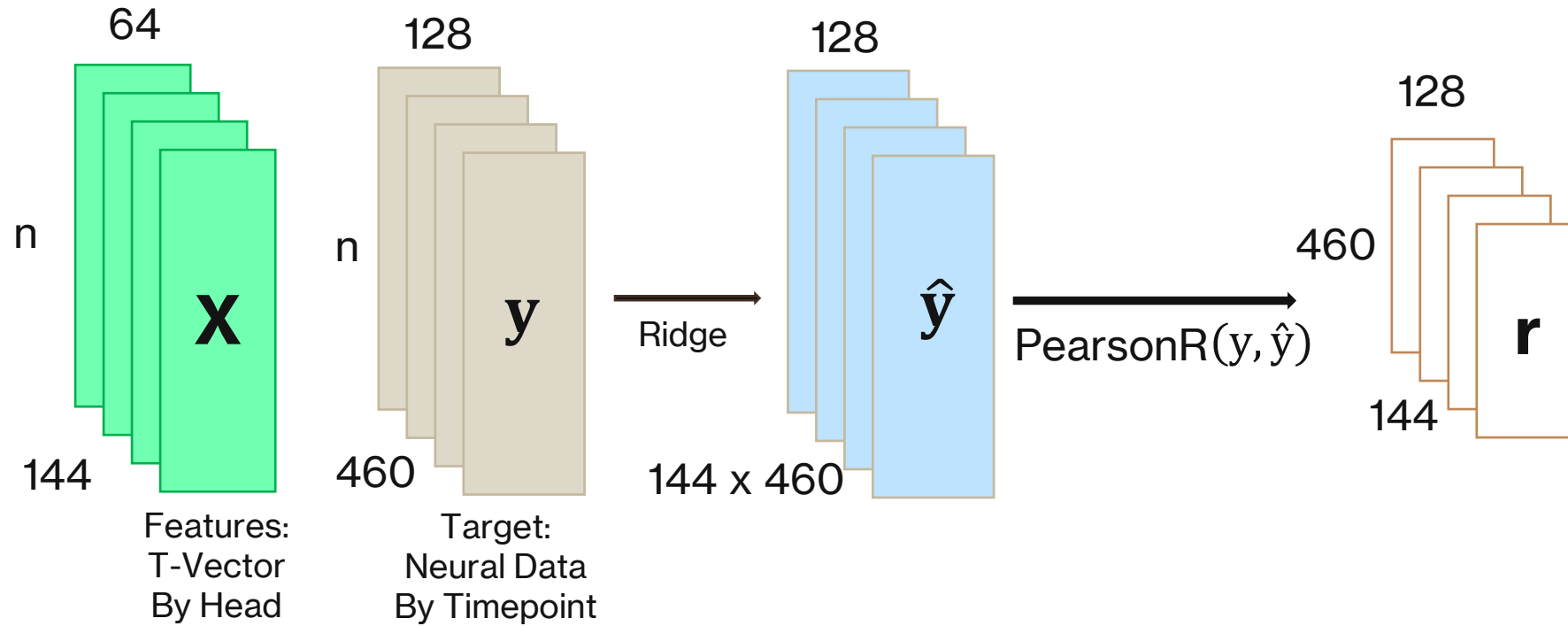
Transformation Vector Encoding Model



Embeddings vs Transformation Vectors



Transformation Vector Encoding Model



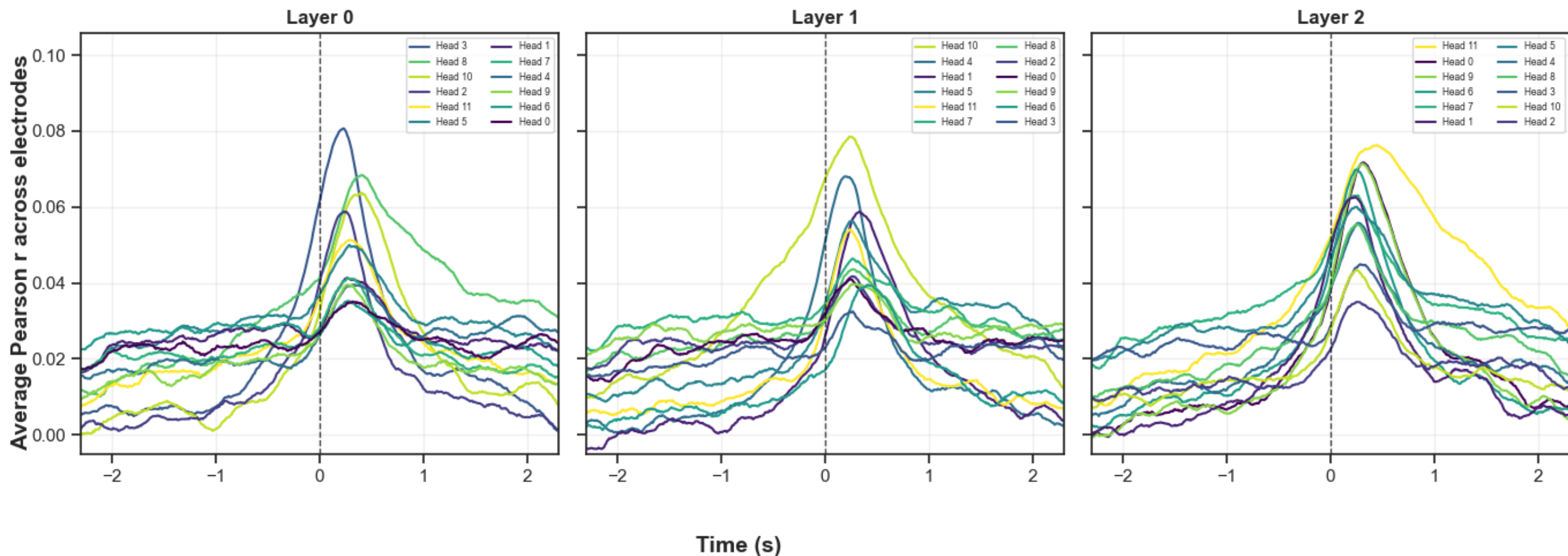
DIFFERENT ATTENTION HEADS DISPLAY DISPARATE BEHAVIOR

Finding 1

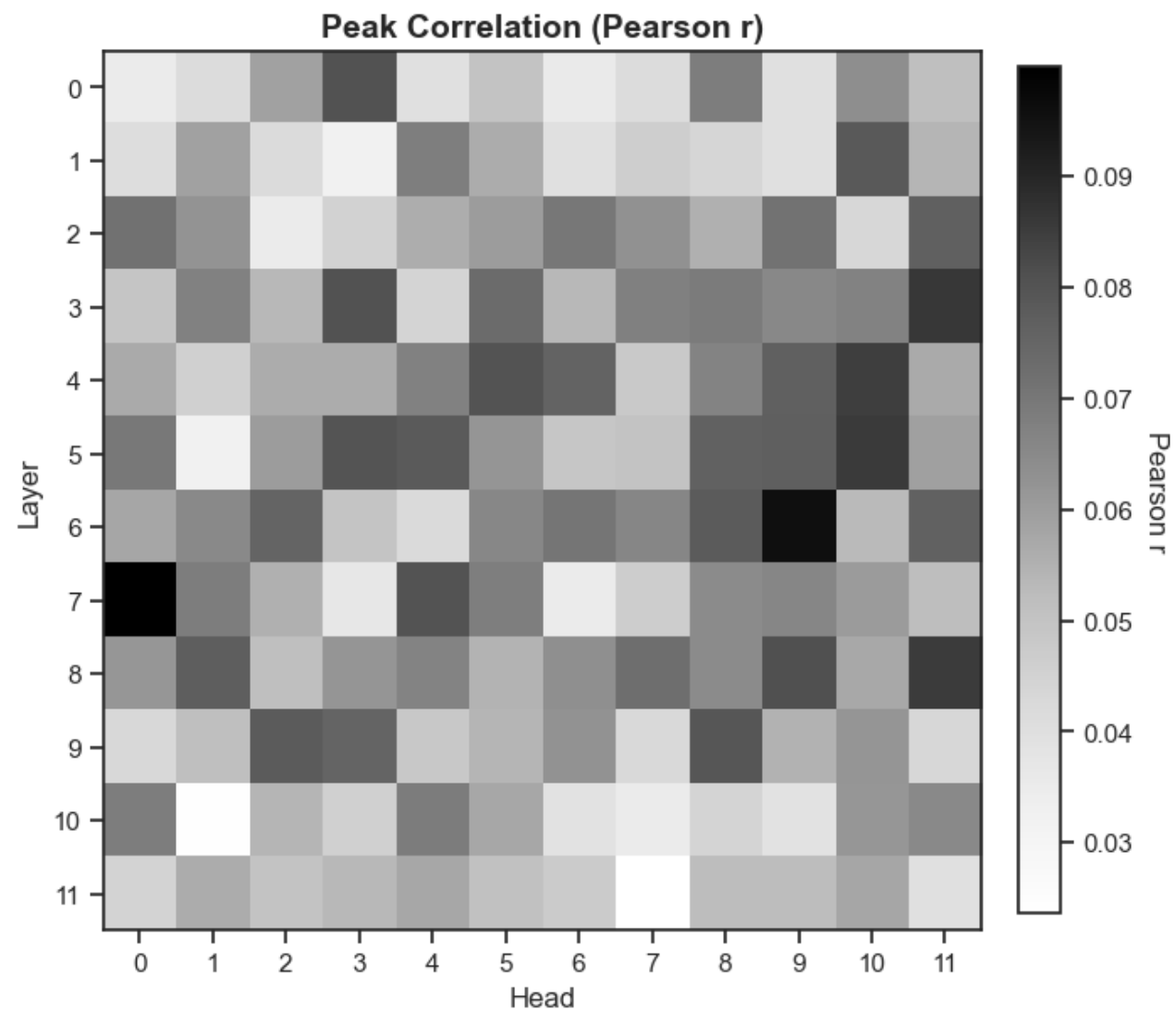
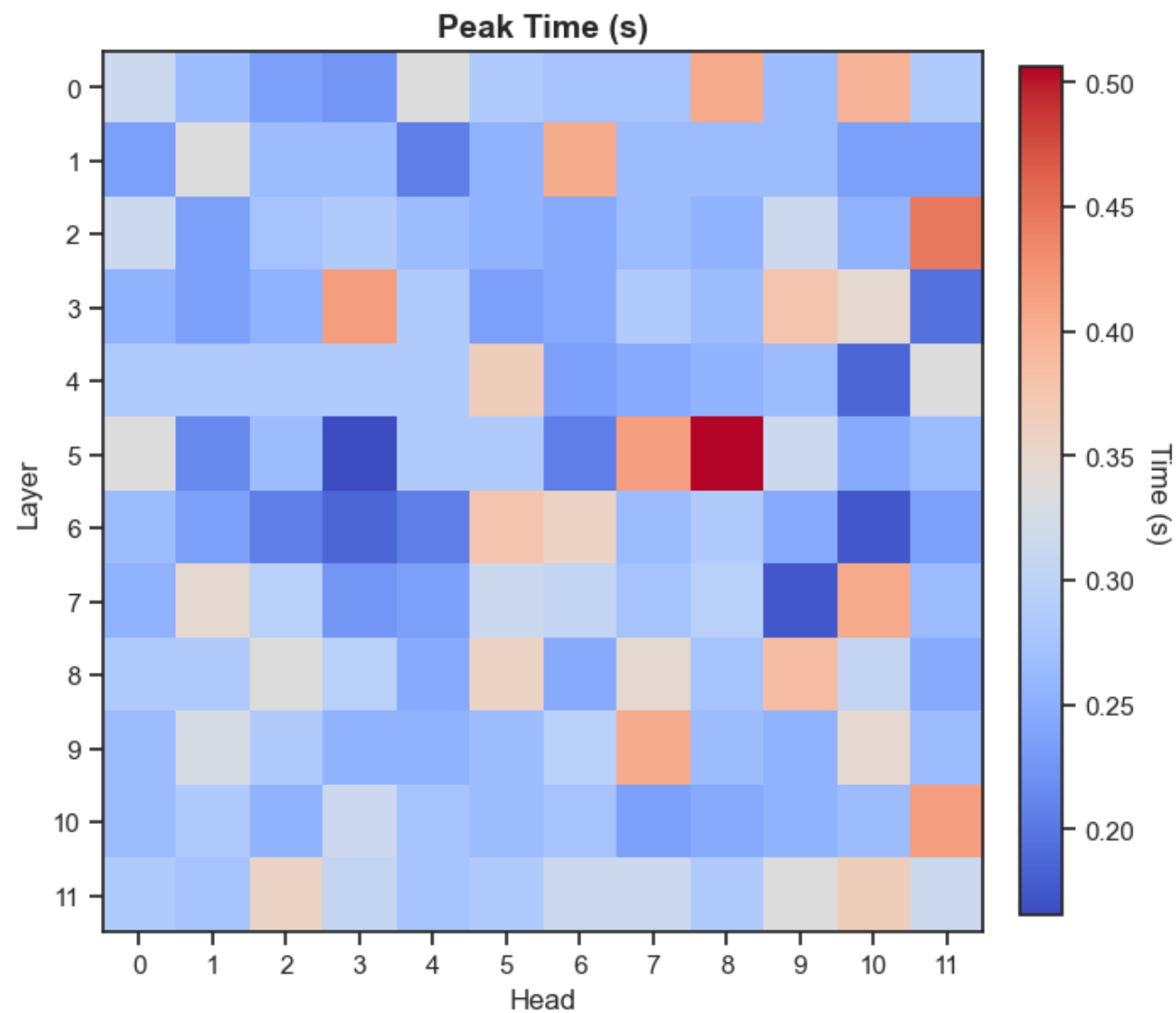


Transformation Vector By Head

Average Pearson r by Layer and Head (Transformations)

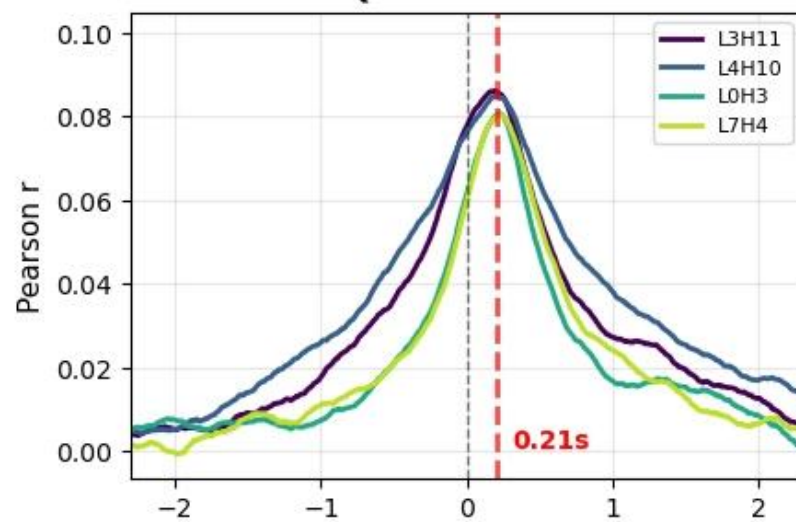


Transformation Vector By Head

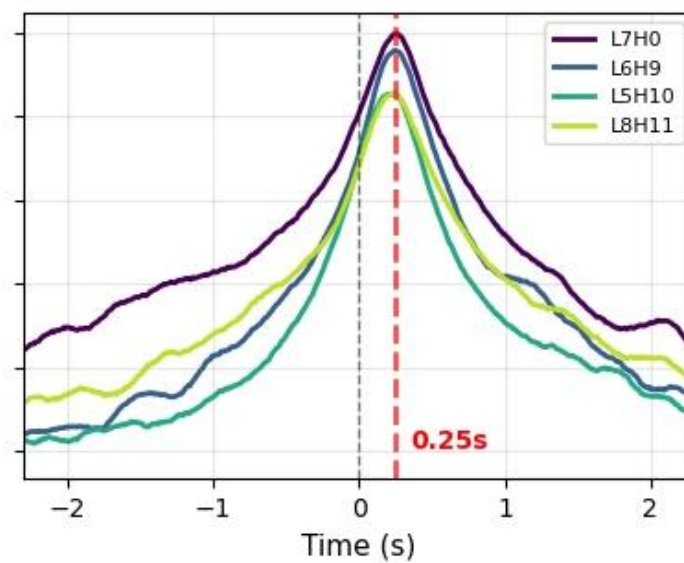


Head Categorization by Peak Timing (Transformations)

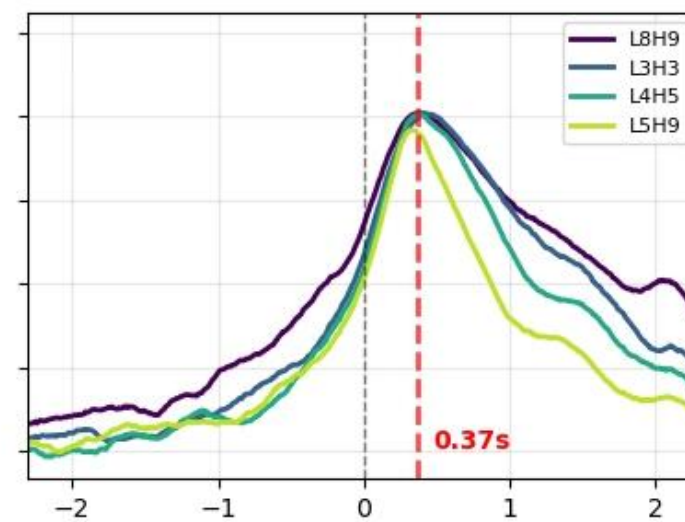
Quick Heads



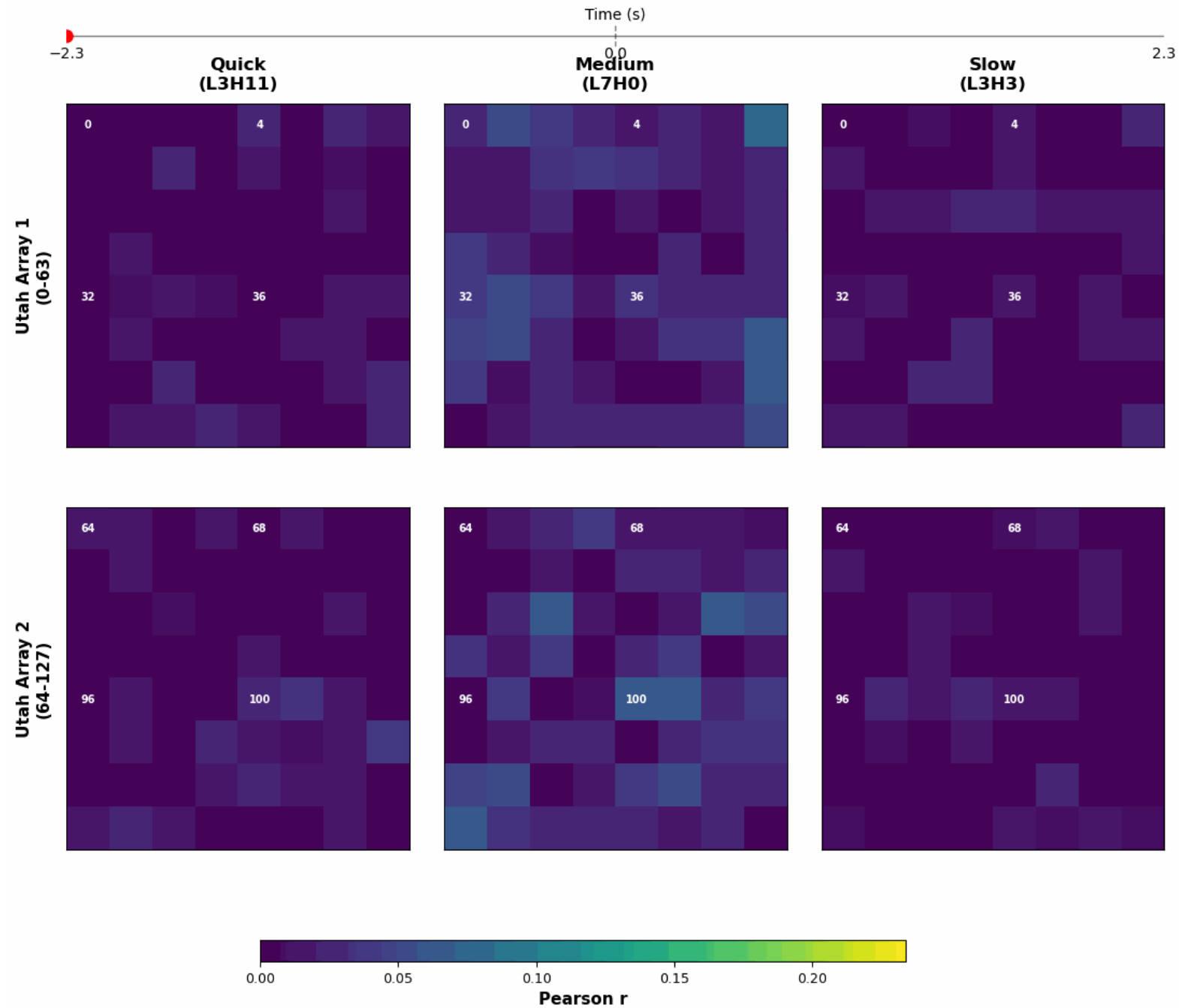
Medium Heads



Slow Heads

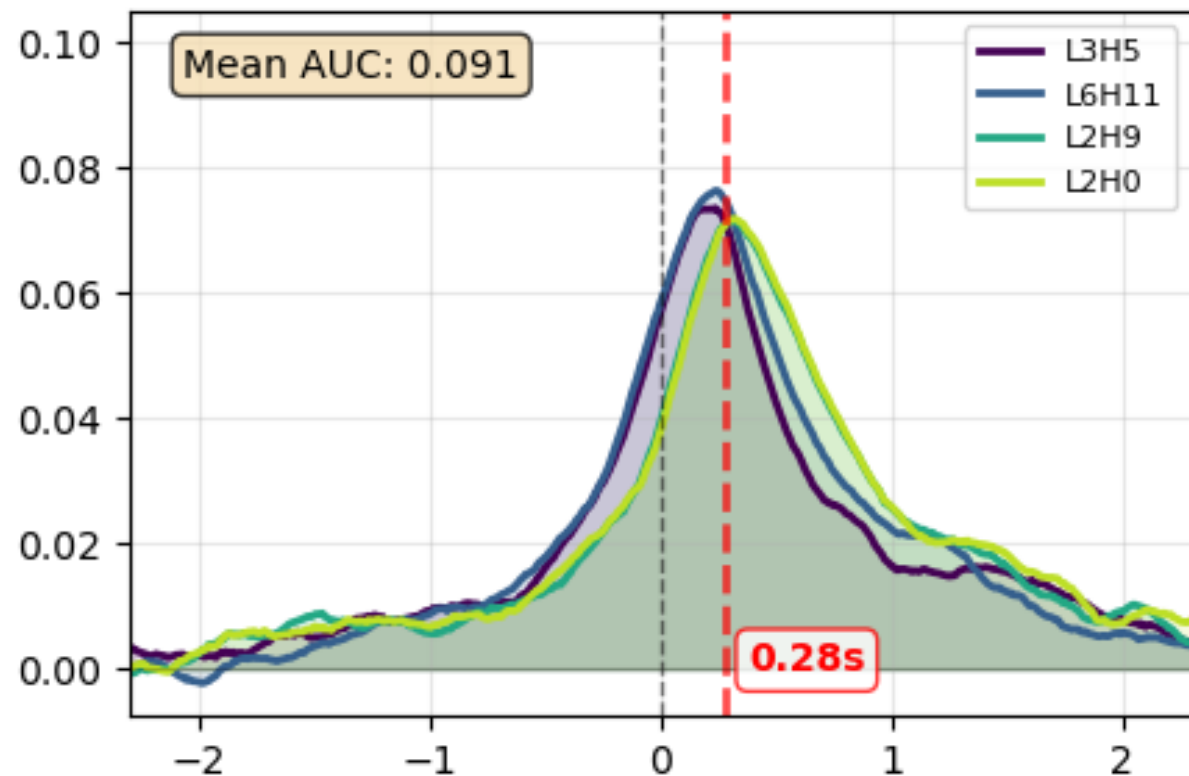


Best Head per Category - Time: -2.30s

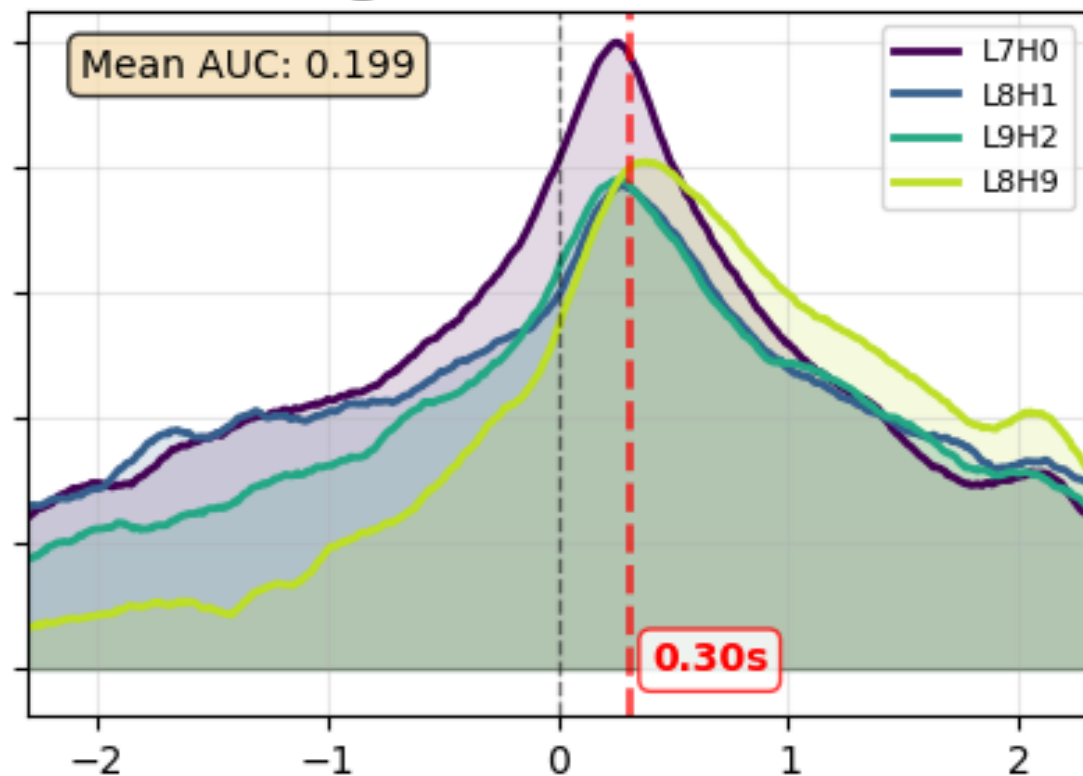


Head Categorization by Duration (AUC) (Transformations)

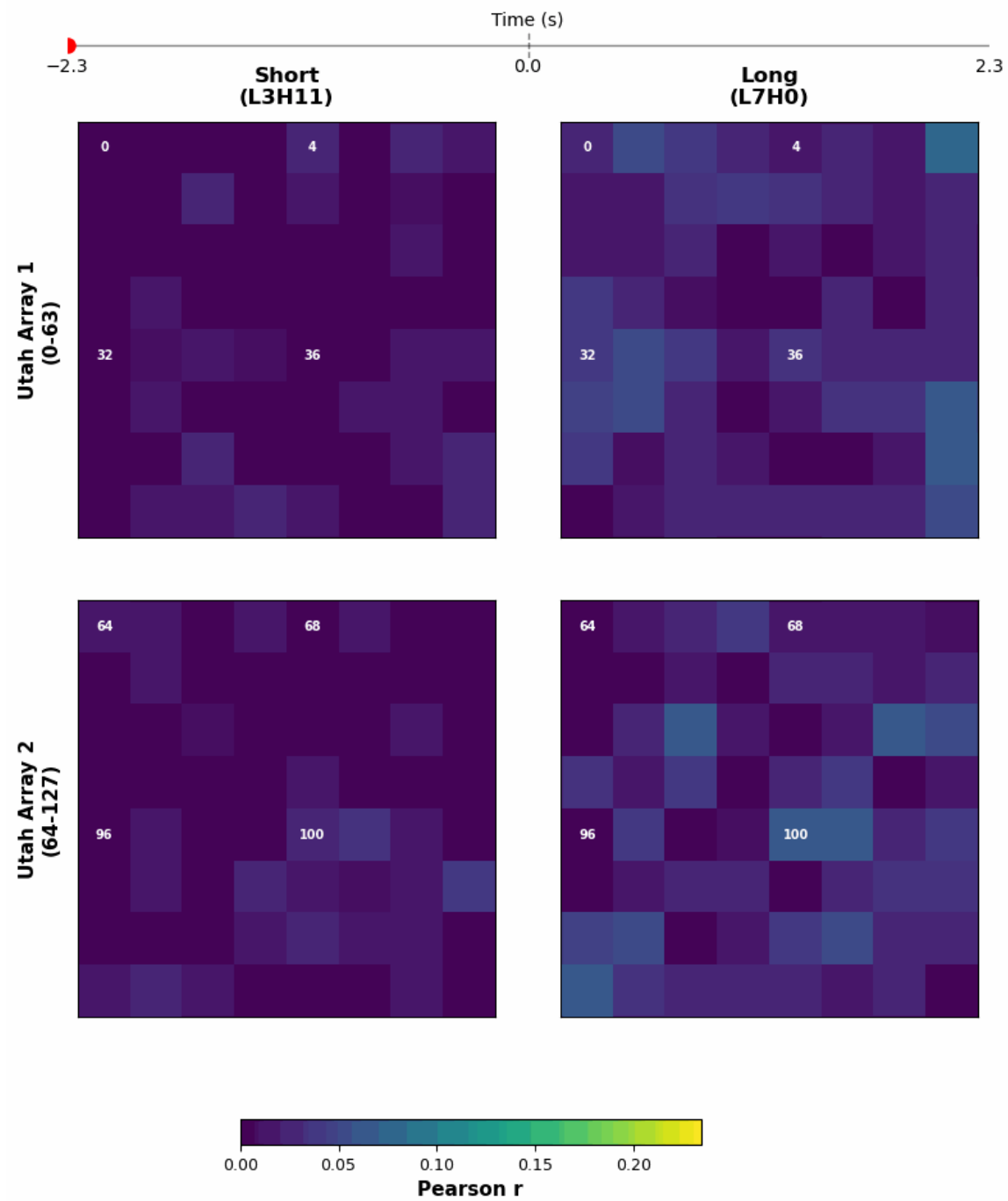
Short Duration Heads



Long Duration Heads

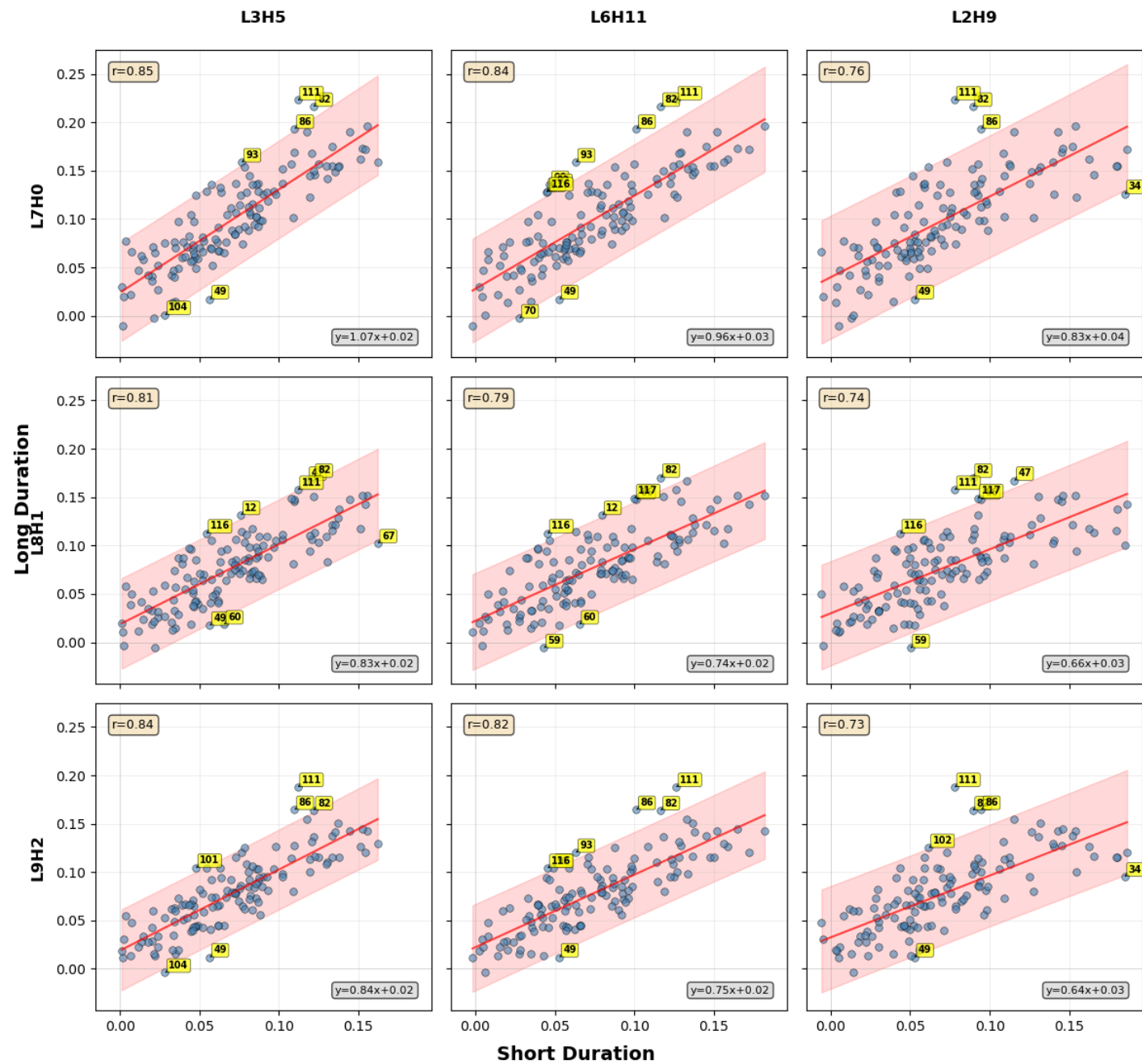


Best Head per Category - Time: -2.30s



Electrode Correlation Comparison at 0.29s (Transformations)

95% PI





TEMPORAL SHIFTS IN ENCODING PERFORMANCE IS CORRELATED WITH DEPENDENCY RELATIONS

Finding 2

What is a Dependency Relation?

A way to characterize asymmetric links between words in a sentence to define syntactic structure.

Some common examples are:

Nominal Subject (nsubj): a non-clausal constituent in the subject position of an active verb

The **dog** barked. (Offset = 1)

Prepositional Modifier (prep): any prepositional phrase that modifies the meaning of its head.

The cat sat **on** the mat. (Offset = -1)

Adverbial Modifier (advmod): an adverb (phrase) that modifies the meaning of another word.

The dog barked **loudly**. (Offset= -1)

The **very** tall tree. (Offset = 1)

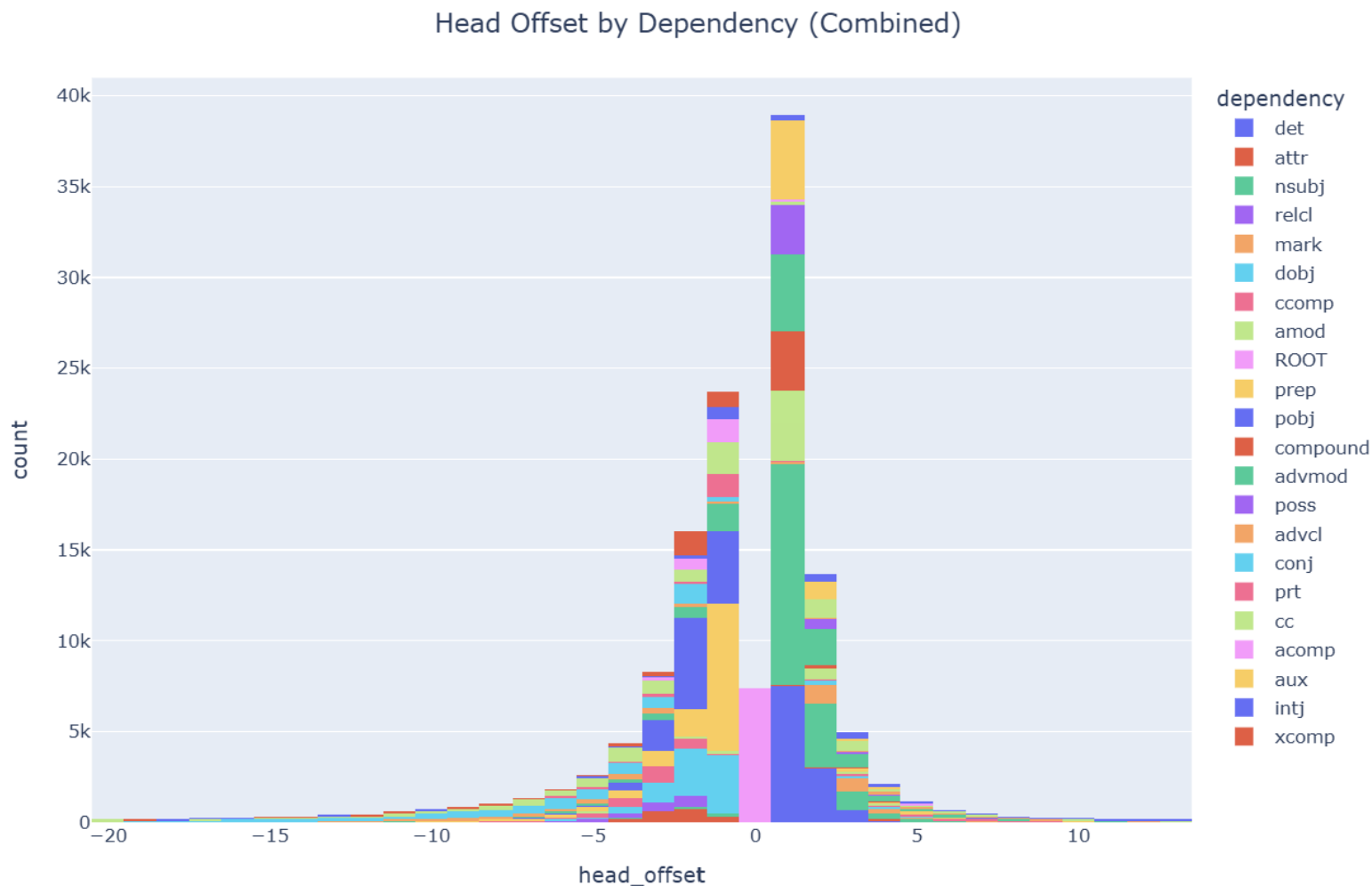
Why do we care about dependencies?

There may be shared specialization between the earlier layers of BERT and the IFG as they have both been to process syntactic relationships between words.

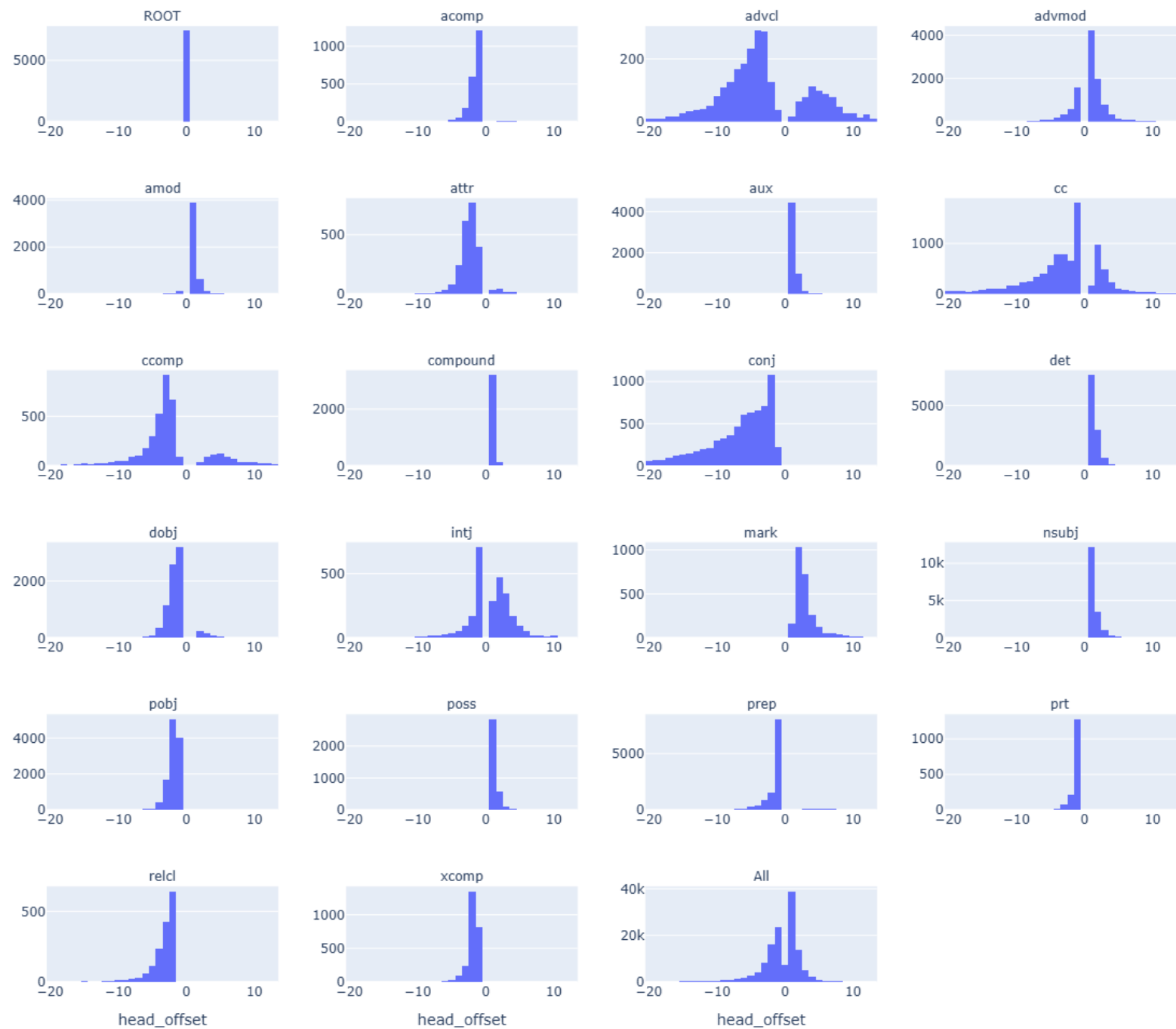
Managing long range dependencies is one of the greatest strengths of transformer-based language models

Dependencies in BERT

We limited our analysis to 22 most common dependencies in the corpus (occurred >1500 times).
Most analysis were limited to samples with offsets ± 5

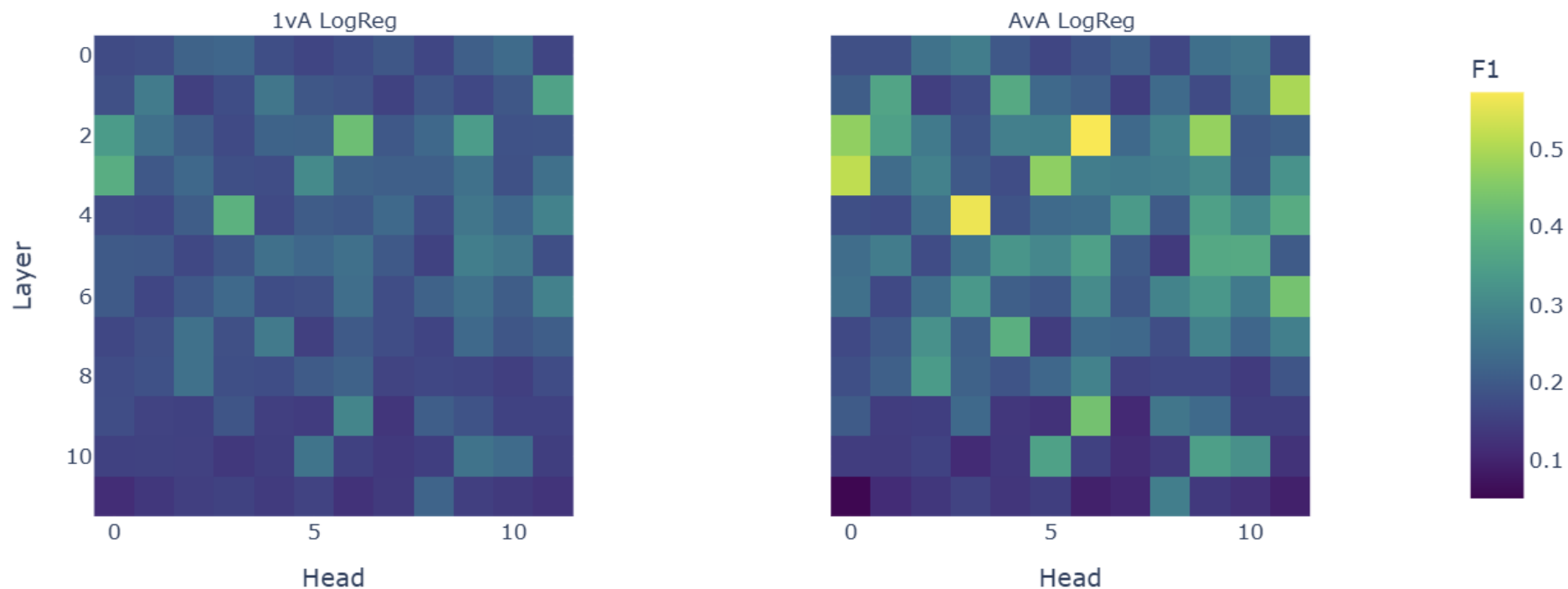


Head Offset by Dependency



Predicting Dependency Type From T-Vector

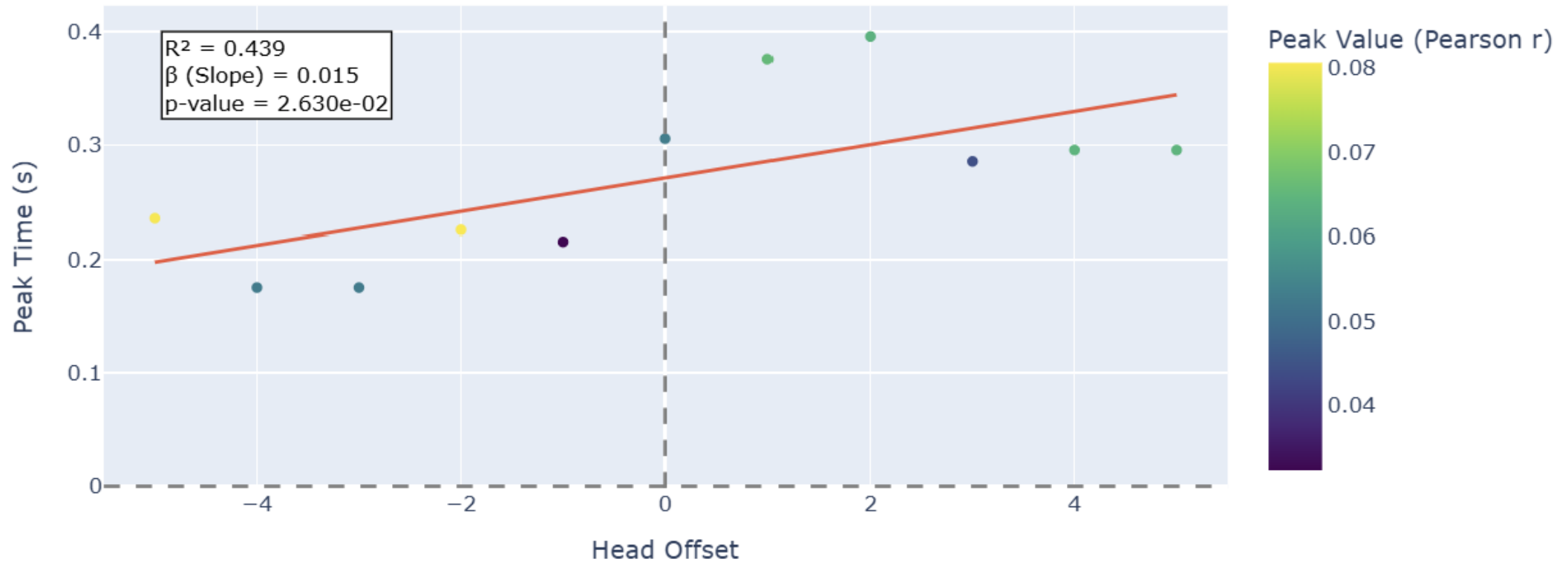
Comparison of Attention Heads (1vA vs AvA)



Some heads are generalists, some heads are more specialized
Some dependencies are a lot easier to predict from many different heads than others

Specialization Correlates with Peak Time

Peak Time vs Head Offset (Most Specialized Head per Offset)



Plans to look in seconds

Specialization Correlates with Peak Time

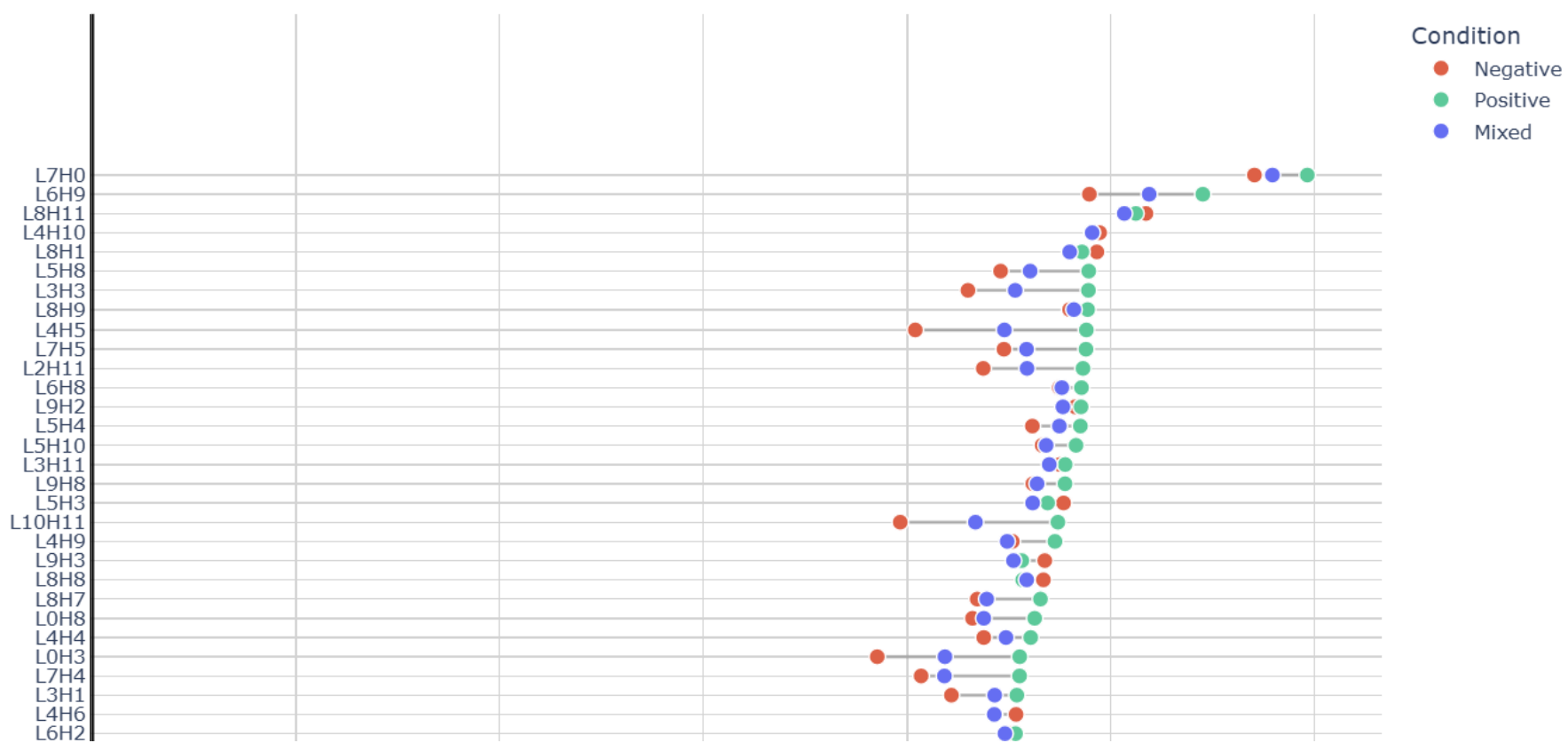
All Attention Heads: Peak Time vs Median Offset of Best Feature (Standardized F1 score)



Underlying Distribution Confounding

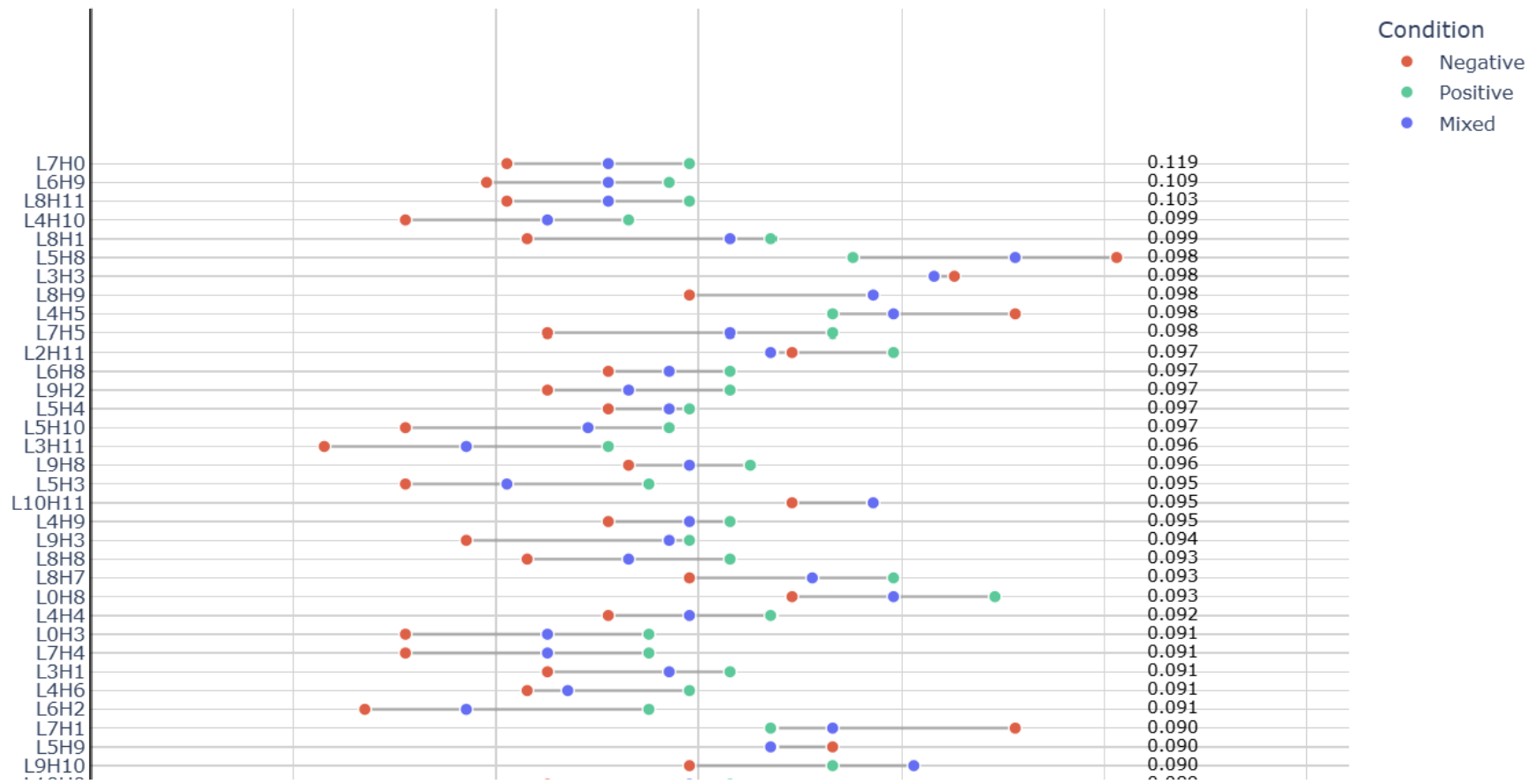
Comparing Peak Times on Positive Offset Words vs Negative Offset Words

Peak Pearson r by Attention Head (Dumbbell Plot)

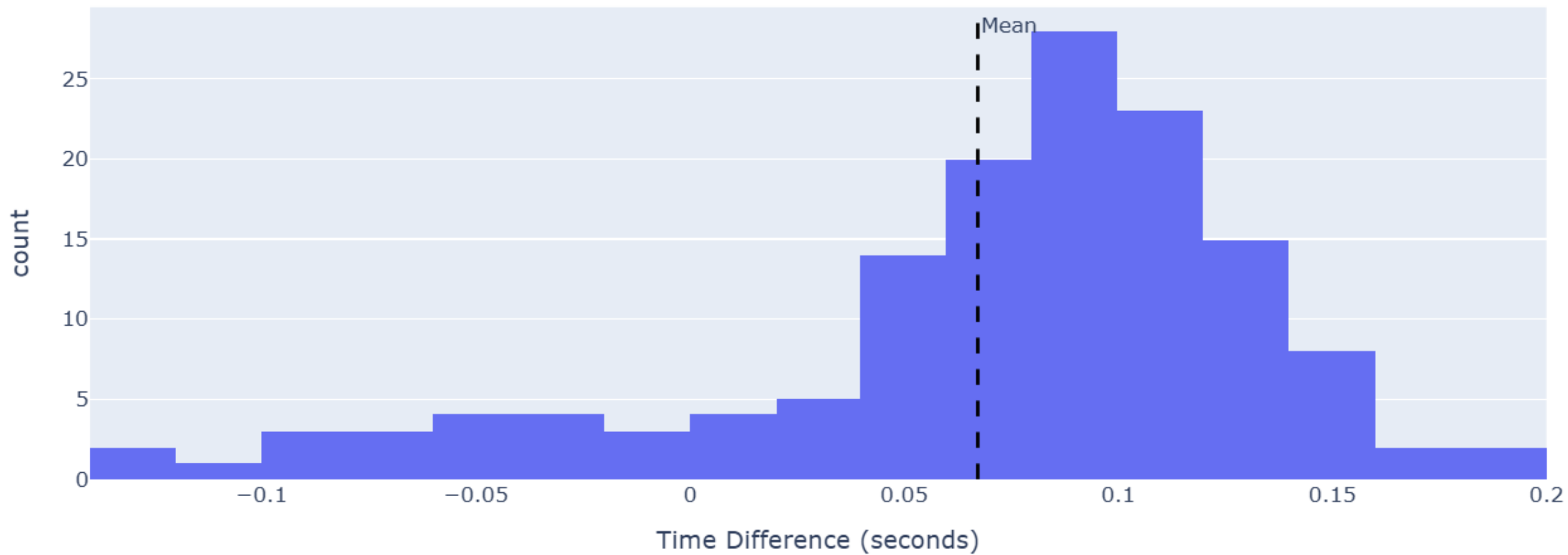


Comparing Peak Times on Positive Offset Words vs Negative Offset Words

Peak Time Range by Attention Head (Sorted by Peak Pearson r)



Distribution of Peak Time Distances (Positive - Negative)



Next Steps

- Look at specific electrodes, particularly specifically tuned to dependencies
 - Disentangle from averages
- Run more dependency specific analyses

References

- 1) Fedorenko, E., Duncan, J., & Kanwisher, N. (2012). Language-selective and domain-general regions lie side by side within Broca's area. *Current biology : CB*, 22(21), 2059–2062. <https://doi.org/10.1016/j.cub.2012.09.011>
- 2) Ishkhanyan, B., Michel Lange, V., Boye, K., Mogensen, J., Karabanov, A., Hartwigsen, G., & Siebner, H. R. (2020). Anterior and Posterior Left Inferior Frontal Gyrus Contribute to the Implementation of Grammatical Determiners During Language Production. *Frontiers in psychology*, 11, 685. <https://doi.org/10.3389/fpsyg.2020.00685>
- 3) Klaus, J., & Hartwigsen, G. (2019). Dissociating semantic and phonological contributions of the left inferior frontal gyrus to language production. *Human brain mapping*, 40(11), 3279–3287. <https://doi.org/10.1002/hbm.24597>